main.c

Run ▶

```c
#include <stdio.h>
#include <ctype.h>
#include <string.h>

int main()
{
    char expr[100], op1, op2, oper;
    char temp = '1';
    int i;

    printf("Enter an expression (e.g., a+b*c): ");
    scanf("%99s", expr);

    printf("\nThree Address Code:\n");

    for (i = 0; expr[i] != '\0'; i++)
    {
        if (expr[i] == '*' || expr[i] == '/')
        {
            op1 = expr[i - 1];
            oper = expr[i];
            op2 = expr[i + 1];

            printf("t%c = %c %c %c\n",
                   temp, op1, oper, op2);

            expr[i - 1] = 't';
```

Output

```
Enter an expression (e.g., a+b*c): a+b*c

Three Address Code:
t1 = b * c
t2 = a + t
```

main.c

Run ▶

```
                temp, op1, oper, op2);

            expr[i - 1] = 't';
            expr[i] = temp;
            expr[i + 1] = ' ';
            temp++;
        }
    }

    for (i = 0; expr[i] != '\0'; i++)
    {
        if (expr[i] == '+' || expr[i] == '-')
        {
            op1 = expr[i - 1];
            oper = expr[i];
            op2 = expr[i + 1];

            printf("t%c = %c %c %c\n",
                   temp, op1, oper, op2);

            temp++;
        }
    }

    return 0;
}
```

Output

```
Enter an expression (e.g., a+b*c): a+b*c

Three Address Code:
t1 = b * c
t2 = a + t
```